Jose de Jesus Herrera Ledon

AI Systems Engineer
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Professional Summary

AI systems and machine learning engineer with 10+ years of experience designing software, data platforms, and high-performance computing solutions. Currently working in S&P Global's Enterprise Data Organization (EDO), building information-extraction agents for energy data and oil wells, plus processes that connect specialized agents into useful extraction workflows.

Core focus: Agentic AI, information extraction, energy data, machine learning, data engineering, Python, PyTorch, TorchRec, SYCL/XPU, cloud platforms, and technical leadership.

Experience

S&P Global — Enterprise Data Organization (EDO) **Mexico City, Mexico**
Machine Learning Data Engineer *11/15/2025–Present*

- Develop information-extraction agents for energy data, including oil-well information.
- Create processes that connect specialized agents to extract and structure energy data.
- Build agent-based workflows that turn complex energy information into usable insights.

Intel **Remote**
AI Software Engineer *08/2022–09/2025*

- Developed and optimized AI frameworks for high-performance computing in TorchRec, including DLRM2 and DLRM3 models.
- Implemented matrix multiplication optimizations for XPU workloads with FBGEMM.
- Led a Cloud Data Connector library and deployed a PyPI package supporting AWS, Azure, and GCP.
- Developed a VS Code extension for Intel AI tools and mentored engineers on SYCL and XPU/GPU optimization.
- Contributed to open-source software and cross-team engineering initiatives.

UMBA **Mexico City**
Senior Software Engineer *07/2021–08/2022*

- Developed and maintained web applications using Flask and TypeScript.
- Led the customer experience team and built tools for engagement, analytics, and reporting.
- Created RESTful APIs for transactions and data processing.
- Collaborated with financial and engineering teams to improve compliance and efficiency.

Intel **Guadalajara, Mexico**
Senior Software Engineer *11/2019–07/2021*

- Developed software for quality and business intelligence.
- Designed large-scale data pipelines and Airflow workflows.
- Designed and migrated OLAP databases and built real-time analytics with Django.

CONTPAQi **Guadalajara, Mexico**
Senior Software Engineer *10/2018–07/2019*

- Worked in the Research and Technology department on financial software solutions.
- Created analytics for transactions and user behavior.
- Implemented machine learning and NLP algorithms for data analytics.

HCL Technologies **Guadalajara, Mexico**
Team Lead Software Engineer *08/2016–09/2018*

- Led a team developing enterprise machine learning and analytics applications.
- Introduced Agile practices and built tools for international tariff-table analysis.

IBM **Guadalajara, Mexico**
Software Developer, CI/CD *02/2015–08/2016*

- Implemented CI/CD pipelines with cross-functional engineering teams.
- Developed analytics and dependency-analysis tools for large codebases.

Hitts Technologies **Guadalajara, Mexico**
Software Developer *07/2014–02/2015*

- Developed and maintained enterprise Java applications for IBM.
- Provided L3 support for Rational Jazz Foundation.

Technical Skills

AI / ML: PyTorch, TorchRec, NLP, information extraction, AI agents, machine learning workflows

Data: Energy data, data engineering, data pipelines, PostgreSQL, MySQL, MongoDB, Redis

Systems: SYCL, XPU, FBGEMM, high-performance computing, Linux

Programming: Python, Java, JavaScript, C++

Cloud / Tools: AWS, Azure, GCP, Docker, Kubernetes, Git, Airflow, CI/CD

Web: Django, TypeScript, FastAPI, Flask

Languages: English, Spanish

Education & Certifications

Autonomous University of Mexico **Mexico City**
Computer Engineer *2008–2013*
View institution

MIT Professional Education
Machine Learning From Data To Decisions *2018*
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MIT XPro
Data Science and Big Data Analytics *2019*
View badge

MIT XPro
Quantum Algorithms for Cybersecurity, Chemistry and Optimization *2019*
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Quantum Computing Fundamentals *2019*
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MIT XPro
Introduction to Quantum Computing *2019*
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Intel Gaudi AI Accelerator *2025*
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